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HOME TOP CATALOG CONTESTS GYM PROBLEMSET GROUPS RATING EDU API CALENDAR HELP

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

A. Autocomplete

time limit per test: 1 second
memory limit per test: 256 megabytes

Peter Mount and Ryei are in a very close Scrabble match. In the game, they must form words and earn points according to the rules. However, the Scrabble played in the MaratonaCIn is a little different: in this version, only words from a specific predetermined set can be formed.

Ryei realized he wouldn't be able to beat Peter Mount, as he is a master of the game. So, he decided to change his strategy: before trying to play a word, he would test which ones were more promising, that is, those that could be transformed into other words within the allowed set. To do this, given a word, he needs to know how many words, that are in the set, he can create, using the given word as a prefix.

Since Ryei is focused on the game, he asked for your help to create a program to assist him in this task. Can you help him?

Input

- The first line contains two integers, N and Q $(1 \le N, Q \le 10^5)$.
- The next N lines contain one word each, formed only by lowercase letters of the English alphabet.
- The following Q lines contain one word each, formed only by lowercase letters of the English alphabet.
- The sum of the lengths of all the provided words does not exceed 10^6.

Output

- For each of the Q queries, print a single integer on a new line, representing the number of words from the original list that have the query word as a prefix.

Example

input Copy

```
4 5
a
ab
abc
abcd
a
aa
ab
abc
abcd
```

output Copy

```
4
0
3
2
1
```

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→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msy:
Choose file: Choose File no file selected
Submit

→ Contest materials

- Statements #1 (pt)
- Statements (PDF) (en)

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